KING SALMAN INTERNATIONAL UNIVERSITY

Field Engineering Sciences

All Programs

Lecture : Binary Representation, ASCII Codes and Number Systems

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Learning outcomes

By the end of the lecture:

Number systems

Explain decimal, binary, octal and hexadecimal; convert between bases.

Character codes

Explain ASCII and distinguish it from Unicode.

Why computers use binary

Digital circuits can reliably represent two states.

Two states

ON/OFF; HIGH/LOW voltage; TRUE/FALSE. These states can be represented by 1 and 0.

Binary representation

Computers represent numbers, text, images, sound and instructions as bit patterns. Meaning depends on how the bits are interpreted.

A bit pattern alone does not tell us the data type.

Key terms: bit, byte and word

Terms describing digital information.

Bit and byte

Bit: one binary digit, 0 or 1.

Nibble: 4 bits.

Byte: 8 bits.

KB, MB and GB describe larger amounts of data.

Word

A word is the natural data size a processor handles as a unit. Word size depends on processor design; common examples are 32-bit and 64-bit architectures.

What is a number system?

A number system uses symbols and rules to represent quantities.

Base or radix

Base tells how many digits are used.

Decimal: base 10

Binary: base 2

Octal: base 8

Hexadecimal: base 16

Place value

In a positional system, a digit's value depends on the digit, its position and the base.

Decimal system (base 10)

Decimal uses digits 0 to 9.

Place values

$4{,}582 = 4\times1{,}000 + 5\times100 + 8\times10 + 2\times1$

$= 4{,}000 + 500 + 80 + 2$

Expanded form

$4{,}582 = 4\times10^3 + 5\times10^2 + 8\times10^1 + 2\times10^0$. Positions are powers of 10, starting at 10^0 on the right.

Binary system (base 2)

Binary uses only 0 and 1.

Binary place values

From right to left: … 32, 16, 8, 4, 2, 1. These are powers of 2.

Example: 1011_2

$1×8 + 0×4 + 1×2 + 1×1 = 11_{10}$. Subscripts show the base.

Convert binary to decimal

Multiply each bit by its place value, then add.

Worked example: 11010_2

Place values: 16, 8, 4, 2, 1

$1\times16 + 1\times8 + 0\times4 + 1\times2 + 0\times1 = 26_{10}$

Reliable method

1. Write place values from right to left.
2. Multiply each bit by its place value.
3. Add values where the bit is 1.
4. Label the answer as decimal.

Convert decimal to binary

Use repeated division by 2 and record remainders.

Worked example: 13_{10}

13÷2 = 6 r1

6÷2 = 3 r0

3÷2 = 1 r1

1÷2 = 0 r1

Read bottom to top: 1101_2

Why bottom to top?

The first remainder is the least significant bit (rightmost). The last remainder becomes the most significant bit (leftmost).

Decimal to binary: place values

Build the number using powers of 2.

Worked example: 22_{10}

22 = 16 + 4 + 2

Place values: 16 8 4 2 1

Bits: 1 0 1 1 0

So $22_{10} = 10110_2$

Check the result

$10110_2 = 16 + 0 + 4 + 2 + 0 = 22_{10}$. Always convert back to check.

Binary counting

Binary uses the same counting principle as decimal, but only 0 and 1.

Count from 0 to 7

Decimal | Binary
0 | 000
1 | 001
2 | 010
3 | 011
4 | 100
5 | 101
6 | 110
7 | 111

Pattern

When a bit changes from 1 to 0, the next position receives a carry.
Example: $0111_2 + 1 = 1000_2$.

Binary addition

Learn the four basic rules.

Addition rules

0+0=0

0+1=1

1+0=1

$1+1=10_2$

For 1+1, write 0 and carry 1.

Worked example

1011_2

$+ 0011_2$

1110_2

Check: $11_{10} + 3_{10} = 14_{10}$.

Binary fractions

Bits to the right of the binary point represent fractions.

Fraction place values

$2^{-1} = 1/2 = 0.5$

$2^{-2} = 1/4 = 0.25$

$2^{-3} = 1/8 = 0.125$

Example: 10.1_2

$1\times2^{1} + 0\times2^{0} + 1\times2^{-1} = 2.5_{10}$. Some decimal fractions repeat in binary.

Octal system (base 8)

Octal uses digits 0 to 7.

Place values

From right to left: … 512, 64, 8, 1. These are powers of 8.

Example: 157_8

$1\times8^2 + 5\times8^1 + 7\times8^0 = 64 + 40 + 7 = 111_{10}$.

Hexadecimal system (base 16)

Hexadecimal uses 16 symbols: 0–9 and A–F.

Hex digits

A=10, B=11, C=12, D=13, E=14, F=15.

Place values

From right to left: … 4096, 256, 16, 1. These are powers of 16.

Why use hexadecimal?

Hexadecimal is a compact way to write long binary patterns.

Example

Binary: $1111\ 1010_2$

Hexadecimal: FA_{16}

Each group of four bits maps to one hex digit.

Applications

Memory addresses and debugging; web colour codes (e.g. #1A73E8); embedded systems; digital electronics.

Binary and hexadecimal conversion

Group binary digits into sets of four.

Binary to hex

$1110\ 0101_2$

1110 = E; 0101 = 5

So $11100101_2 = E5_{16}$.

Hex to binary

$3A_{16}$: 3=0011; A=1010. So $3A_{16} = 00111010_2$. Leading zeros can show complete groups.

Binary and octal conversion

Group binary digits into sets of three.

Binary to octal

$110\ 101_2$

110 = 6; 101 = 5

So $110101_2 = 65_8$.

Octal to binary

47_8: 4=100; 7=111. So $47_8 = 100111_2$.

Compare four number systems

The same quantity can be represented in different bases.

Equivalent values

Decimal: 26_{10}

Binary: 11010_{2}

Octal: 32_{8}

Hexadecimal: $1A_{16}$

Important rule

The value stays the same; only the representation changes. Use a base label when comparing systems.

What is character encoding?

Computers store text as numbers, so a shared code is needed.

Definition

Character encoding maps characters—letters, digits and punctuation—to numeric code values that can be stored and processed.

Why standards matter

If systems interpret the same code differently, text may appear incorrectly. A shared encoding helps systems exchange text.

ASCII codes

ASCII means American Standard Code for Information Interchange.

Definition

ASCII is a character-coding standard assigning numeric codes to English letters, digits, punctuation and control characters.

Original ASCII range

Standard ASCII uses 7 bits, giving 128 values: 0–127. Codes 0–31 and 127 are control codes; printable characters include letters, digits and punctuation.

ASCII examples

Each standard character has a numeric code.

Selected characters

Space = 32

0 = 48; 1 = 49

A = 65; B = 66; Z = 90

a = 97; b = 98; z = 122

Patterns

Digits 0–9 are consecutive codes. A–Z are consecutive. a–z are consecutive. Character “5” is not the same as numeric value 5.

ASCII in binary and hexadecimal

One code can be written in different number systems.

Example: A

Character A
Decimal ASCII: 65
Binary 7-bit: 1000001
8-bit display: 01000001
Hex: 41_{16}

Example: a

Character a
Decimal ASCII: 97
Binary 7-bit: 1100001
8-bit display: 01100001
Hex: 61_{16}

Control and printable characters

Not every ASCII code creates a visible symbol.

Printable characters

Letters: A, b, z

Digits: 0, 5, 9

Punctuation: ! ? , .

Space is a character even though it looks blank.

Control characters

Control codes were designed to control text devices or communication.
Examples: NUL (null), LF (line feed), CR (carriage return), ESC (escape).

ASCII and Unicode

ASCII cannot represent every writing system.

ASCII limitation

Standard ASCII covers a limited set, mainly English text and control functions. It does not include most Arabic letters or many other writing systems.

Unicode

Unicode assigns code points across many writing systems and symbols. UTF-8 is a widely used Unicode encoding; ASCII characters use the same byte values in UTF-8.

How text becomes stored data

A message is stored as a sequence of encoded character values.

Example: OK

O → ASCII 79 → binary 01001111

K → ASCII 75 → binary 01001011

The stored byte sequence represents “OK”.

Engineering example

A serial device may send characters such as TEMP=25. The receiver must interpret the encoding correctly before processing the text.

Unsigned binary integers

With n bits, an unsigned number represents 0 to 2^n-1.

Ranges

4 bits: 0–15

8 bits: 0–255

16 bits: 0–65,535

More bits allow more possible values.

Why it matters

Sensor readings and counters must fit the chosen number of bits. If a value exceeds the range, overflow may occur.

Signed integers: introduction

Computers also need ways to represent negative numbers.

Sign-and-magnitude

One bit indicates the sign and the remaining bits represent magnitude. This is a useful concept but not the usual modern integer format.

Two’s complement

For n-bit two’s-complement integers, range is -2^{n-1} to $2^{n-1}-1$. For 8 bits: −128 to +127.

Binary representation in engineering

Digital systems encode physical information as bit patterns.

Sensors and control

A sensor voltage is sampled and converted to a digital number by an ADC. A controller processes the number; the result may be displayed, stored or sent to an actuator.

Communication and storage

Communication links send bits representing data. Files store encoded numbers, text, images or sound. Software interprets bits according to a defined format.

Common mistakes

Check the base and the meaning of the bits.

Number-system mistakes

Reading binary as decimal; forgetting place values; reading remainders top-to-bottom; using 3-bit groups for hex or 4-bit groups for octal. Correct: 4 for hex, 3 for octal.

ASCII mistakes

Confusing character code with numeric value; assuming standard ASCII includes Arabic or all accented letters; assuming every encoding always uses one byte.

30-minute guided practice

Work alone first, then compare answers with a partner.

Part A — number systems (15 min)

1. Convert 101101_2 to decimal.
2. Convert 37_{10} to binary.
3. Convert $7B_{16}$ to decimal.
4. Convert 110111_2 to octal.
5. Convert 10101111_2 to hexadecimal.

Part B — ASCII and concepts (15 min)

6. Give decimal ASCII codes for A and a.
7. Write ASCII A as 8-bit binary.
8. Explain one difference between ASCII and Unicode.
9. How many unsigned values can 8 bits represent?
10. Give one engineering use of binary.

Practice answers — number systems

Use the answers to check your method.

Questions 1–5

1. $101101_2 = 45_{10}$
2. $37_{10} = 100101_2$
3. $7B_{16} = 7 \times 16 + 11 = 123_{10}$
4. $110111_2 = 67_8$
5. $10101111_2 = AF_{16}$

Method reminders

Binary→decimal: add active powers of 2. Decimal→binary: repeated division or powers of 2. Hex→decimal: powers of 16. Binary→octal: groups of 3. Binary→hex: groups of 4.

Practice answers — ASCII

Review any answer you missed.

Questions 6–10

6. A=65; a=97.

7. A = 01000001 (8-bit).

8. ASCII is a limited 7-bit standard; Unicode covers many writing systems.

9. 2^8 = 256 values (0–255).

10. Examples: sensor readings, embedded controllers, communication or file storage.

Self-check

Can you explain each answer without reading the slide? If not, return to the worked example and repeat the steps.

Quick quiz (1)

Choose the best answer.

Questions 1–4

1. Base of binary? A 2 B 8 C 10 D 16
2. Bits in a byte? A 4 B 8 C 16 D 32
3. 1010_2 in decimal? A 8 B 9 C 10 D 12
4. Hex digits after 9? A G–L B A–F C 0–5 D X–Z

Questions 5–8

5. Values with 7 bits? A 7 B 64 C 128 D 256
6. Standard ASCII bits? A 4 B 7 C 8 D 16
7. ASCII 65? A a B A C 0 D Space
8. Unicode encoding? A UTF-8 B GHz C MIPS D OCR

Quick quiz (2)

Choose the best answer.

Questions 9–11

9. 1111_2 in decimal? A 7 B 8 C 14 D 15

10. Binary groups for hex? A 2 B 3 C 4 D 8

11. Correct statement? A ASCII covers all scripts. B ASCII maps characters to codes. C Unicode is CPU speed. D Bit is 8 bytes.

Questions 12–15

12. 25_{10} in binary? A 11001 B 10101 C 11100 D 10011

13. Unsigned values in 8 bits? A 8 B 128 C 255 D 256

14. Digits 0–7? A Binary B Octal C Decimal D Hex

15. Character encoding? A maps characters to numeric codes B increases clock speed C stores only pictures D replaces OS

Quiz answer key

Answers with short explanations.

Questions 1–8

1 A—base 2. 2 B—8 bits per byte. 3 C—8+2=10. 4 B—A–F represent 10–15. 5 C—2^7=128 values. 6 B—standard ASCII is 7-bit. 7 B—ASCII 65 is A. 8 A—UTF-8 encodes Unicode text.

Questions 9–15

9 D—8+4+2+1=15. 10 C—4 bits per hex digit. 11 B—ASCII maps characters to codes. 12 A—25=16+8+1=11001_2. 13 D—2^8=256 values. 14 B—octal uses 0–7. 15 A—encoding maps characters to code values.

Lecture summary

Key ideas to remember.

Number systems

Base determines the digit set and place values. Binary is base 2; decimal base 10; octal base 8; hexadecimal base 16. Convert using place values, repeated division or bit grouping.

Binary and text

Computers represent data using bits. ASCII maps a limited character set to codes. Unicode supports many writing systems. The same bits can have different meanings depending on the format.

Exit ticket: convert 100101_2 to decimal and explain one difference between ASCII and Unicode.

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THANK YOU

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